

????

Hannah Scholz

Born 21st July, 2003 in Bonn, Germany

22nd August, 2024

Master's Thesis Mathematics

Advisor: Dr. Michael Rothgang

Second Advisor: Prof. Dr. Floris van Doorn

MATHEMATISCHES INSTITUT

MATHEMATISCH-NATURWISSENSCHAFTLICHE FAKULTÄT DER
RHEINISCHEN FRIEDRICH-WILHELMS-UNIVERSITÄT BONN

Contents

1	The mathematics of collar neighborhoods	5
1.1	Prerequisites	5
1.2	Basics about collar neighbourhoods	5
	Bibliography	9

1 The mathematics of collar neighborhoods

1.1 Prerequisites

Lemma 1.1.1 (Smooth Urysohn's lemma, in mathlib I think). *Let M be a $\mathcal{C}^k$ -manifold and A and B two disjoint closed sets in M . Then there is a $\mathcal{C}^k$ -function $f: M \rightarrow [0, 1]$ such that f is 0 on A and 1 on B .*

Remark 1.1.2. Again, this requires a definition of manifold that makes it metrizable.

1.2 Basics about collar neighbourhoods

Definition 1.2.1. A *topological manifold* is a second-countable Hausdorff space that is locally Euclidean.

Remark 1.2.2. Apparently, there are some subtleties with collars of non-metrizable manifolds but I assume we don't care. (At least with the above definition our manifolds are always metrizable.)

Definition 1.2.3. Let M and N be two $\mathcal{C}^k$ -manifolds. Let $f: M \rightarrow N$.

- (i) f is a $\mathcal{C}^k$ -diffeomorphism if it is bijective and both f and its inverse functions are k -times continuously differentiable.
- (ii) f is a $\mathcal{C}^k$ -embedding if it is a $\mathcal{C}^k$ -diffeomorphism onto its image.

Definition 1.2.4 (Due to [Con71]). Let M be a $\mathcal{C}^k$ -manifold (with boundary) and let $B \subseteq M$ be closed.

- (i) B is *locally collared* if it is covered by sets $U \subseteq B$ which are open in B such that there are $\mathcal{C}^k$ -embeddings $\varphi_U: \overline{U} \times [0, 1] \rightarrow M$ with
 - a) $\varphi_U^{-1}(B) = \overline{U} \times \{0\}$.
 - b) For all $x \in \overline{U}$, $\varphi_U(x, 0) = x$.
 - c) $\varphi_U(\overline{U} \times [0, 1])$ is closed.
 - d) $\varphi_U(U \times [0, 1])$ is open.
- (ii) B is *collared* if B already fulfils the conditions given above. In this case we call the image of $B \times [0, 1)$ under the corresponding map $\varphi_B: B \times [0, 1] \rightarrow M$ a *collar neighbourhood* of B .

Remark 1.2.5. This definition is a lot more restrictive than other definition which don't consider the closure or the closed interval. I think if this one doesn't work, I can just take the other one.

Remark 1.2.6. For $k = 0$ we can even extend this definition to topological spaces instead of just defining it for topological manifolds.

Theorem 1.2.7 (Due to [Con71]). *Let X be a Hausdorff space and $B \subseteq X$ be a closed subset. If B is locally collared in X , then it is collared.*

Proof. Let $U_1, \dots, U_n$ be a cover of B that witnesses that B is locally collared. Let $\varphi_i: \overline{U_i} \times [0, 1] \rightarrow X$ be the corresponding embeddings. B is normal because it is compact and Hausdorff. By the shrinking lemma (already in mathlib) there is a new open cover $V_1, \dots, V_n$ of B such that $\overline{V_i} \subseteq U_i$ for all $1 \leq i \leq n$. Define

$$X^+ := (X \amalg B \times [-1, 0]) / x \sim (x, 0).$$

Clearly, X^+ is collared. So it suffices to show that there is a homeomorphism $G: X \rightarrow X^+$ s.t. $G(x) = (x, -1)$ for all $x \in B$. Let $\Phi_i: \overline{U_i} \times [-1, 1] \rightarrow X^+$ be defined by

$$\Phi_i(x) = \begin{cases} \phi_i(x) & \text{if } x \in \overline{U_i} \times [0, 1] \\ x & \text{if } x \in \overline{U_i} \times [-1, 0]. \end{cases}$$

We now inductively define maps $f_i: B \rightarrow [0, 1]$ and embeddings $g_i: X \rightarrow X^+$ for all $0 \leq i \leq n$ such that

- (i) $f_i(x) = -1$ if $x \in \bigcup_{j \leq i} \overline{V_j}$
- (ii) $g_i(x) = (x, f_i(x))$ if $x \in B$
- (iii) $g_i(X) = X \cup \{(x, t) \mid t \geq f_i(x)\}$

Then g_n will be the desired homeomorphism G from above. Set f_0 to be constant 0. Then we can choose g_0 to be the inclusion and trivially $g_0(X) = X = X \cup X \times \{0\} = X \cup \{(x, t) \mid t \geq 0\}$.

Assume that g_{i-1} has been defined. Using Urysohn's Lemma, choose a continuous function $\lambda_i: \overline{U_i} \rightarrow [0, 1]$ such that it is 0 on $\overline{U_i} \setminus U_i$ and 1 on $\overline{V_i}$. Given $x \in B$ set $a(x) := f_{i-1}(x)$ and $b(x) := (1 - \lambda_i(x))f_{i-1}(x) + \lambda_i(x)(-1)$. Now let

$$s(x, _): [a(x), 1] \rightarrow [b(x), 1], \quad t \mapsto \frac{b(x) - 1}{a(x) - 1}(t - 1) + 1.$$

Now set

$$\psi_i: \Phi_i^{-1}(g_{i-1}(X)) \rightarrow \overline{U_i} \times [-1, 1], \quad (x, t) \mapsto (x, s(x, t)).$$

This allows us to define

$$\Psi_i: g_{i-1}(X) \rightarrow X^+, \quad x \mapsto \begin{cases} \Phi_i(\psi_i(\Phi_i^{-1}(x))) & \text{if } x \in g_i(X) \cap \Phi_i(\overline{U_i} \times [-1, 1]) \\ x & \text{otherwise} \end{cases}$$

Claim 1: Ψ_i is continuous.

Proof. It suffices to show that ψ_i is continuous. We need to show that $s: B \times [a(x), 1] \rightarrow [b(x), 1]$ is continuous.

Ahhh this is ugly. I somehow would prefer to just get going with this in Lean. $\square$

□

I want to try to separate out some parts here:

Definition 1.2.8. Let X be a $\mathcal{C}^k$ -manifold and $B \subseteq X$ a closed subset. We define the topological space

$$X_B^+ := (X \amalg B \times [-1, 0]) / x \sim (x, 0).$$

Lemma 1.2.9. Let X be an n -dimensional $\mathcal{C}^k$ -manifold and $B \subseteq X$ a closed subset. Then X_B^+ is an n -dimensional $\mathcal{C}^k$ -manifold.

Proof. We first need to show that X_B^+ is a Hausdorff space. I am not sure if this is best done manually or if there is some cool lemma like: a pushout along closed embeddings preserves Hausdorff spaces. (Is this true?)

Then we need to show that X_B^+ is second countable. Again, there should be some general lemma about closed embeddings and pushouts.

Now we need to pick an atlas for X_B^+ . Let $\mathcal{A}_X$ be a $\mathcal{C}^k$ -atlas for X , $\mathcal{A}_B$ be a $\mathcal{C}^k$ -atlas for B and $\mathcal{A}_{[-1,0]}$ be a $\mathcal{C}^k$ -atlas for $[-1, 0]$. We define

$$\begin{aligned} \mathcal{A} := & \{(U, \varphi) \in \mathcal{A}_X \mid U \cap B = \emptyset\} \\ & \cup \{(U \times I, \varphi \times \psi) \mid (U, \varphi) \in \mathcal{A}_B, (I, \psi) \in \mathcal{A}_{[-1,0]}, 0 \notin I\} \\ & \cup \{(U \cup ((U \cap B) \times I), \varphi_I^+) \mid (U, \varphi) \in \mathcal{A}, (I, \psi) \in \mathcal{A}_{[-1,0]}, 0 \in I\} \end{aligned}$$

where we define φ_I^+ as follows. Let (U, φ) and (I, ψ) as above. Set

$$\varphi_I^+ : U \cup ((U \cap B) \times I) \rightarrow \mathbb{R}^n, x \mapsto \begin{cases} \varphi(x) & x \in U \\ (\varphi(y), z) & x = (y, z) \in (U \cap B) \times I \end{cases}$$

For $x \in B$ we have $\varphi(x)_I^+ = \varphi(x) = (\varphi(x), 0) = \varphi_I^+(x, 0)$, so φ_I^+ is continuous.

It seems to me that this might not be $\mathcal{C}^k$ at all. I don't know if you can even glue this without basically already having the vector field available and then one might as well do the other proof. Maybe a local vector field is enough? Not sure :(

Now you would need to verify that this is a $\mathcal{C}^k$ atlas. Possibly using a version of the manifold chart lemma from Lee. I will only try to show that the transition maps are $\mathcal{C}^k$. The rest should be easier. The only interesting intersections are those of two charts of the form $(U \cup ((U \cap B) \times I), \varphi_I^+)$ and $(V \cup ((V \cap B) \times J), \psi_J^+)$. We need to check that the transition map is $\mathcal{C}^k$ on $U \cap V \cap B$. This is clear for all combinations of partial derivatives that do not include the n -th one. □

Lemma 1.2.10. Let X be an n -dimensional $\mathcal{C}^k$ -manifold and $B \subseteq X$ a closed subset. Then X_B^+ is the pushout in the category of $\mathcal{C}^k$ -manifolds.

Remark 1.2.11. It seems that this is not at all trivial and in fact much of the issue with gluing manually along boundaries. I am not sure if we can even guarantee the above lemma. The above lemma would hopefully mean that one never has to look at the explicit construction again.

Definition 1.2.12. Let X be a Hausdorff space and $B \subseteq X$ a closed subset. Let $U \subseteq B$ be an open subset together with a

We would like the following theorem:

Theorem 1.2.13. *Let M be a C^k -manifold (with boundary) and $B \subseteq X$ a closed subset. If B is locally collared in M , then it is collared.*

Then we will need the following lemma:

Lemma 1.2.14. *Let M be an n -dimensional C^k -manifold with boundary. Then ∂M is locally collared.*

Proof. Let $p \in \partial M$. Then p is contained in a regular half-coordinate ball U , i.e. there is a coordinate half-ball $U' \supseteq \bar{U}$ and a coordinate map $\psi: U' \rightarrow \mathbb{H}^n$ such that for some $r' > r > 0$ we have $\psi(U) = B_r(0) \cap \mathbb{H}^n$, $\psi(\bar{U}) = \overline{B_r(0)} \cap \mathbb{H}^n$ and $\psi(U') = B_{r'}(0) \cap \mathbb{H}^n$.

Now pick an open neighbourhood V of $0 \in \partial \mathbb{H}^n$ and $l > 0$ s.t. $V \times [0, l] \subseteq B_r(0)$.

Considering $\bar{V}$ and V as submanifolds of $\mathbb{H}^n$ and $\psi^{-1}(\bar{V}) = \overline{\psi^{-1}(V)}$ and $\psi^{-1}(V)$ as submanifolds. Then

$$\psi|_{\overline{\psi^{-1}(V)}}: \overline{\psi^{-1}(V)} \rightarrow \bar{V}$$

and

$$\psi|_{\psi^{-1}(V)}: \psi^{-1}(V) \rightarrow V$$

are C^k -diffeomorphisms. So it suffices to show that $\psi^{-1}|_{\bar{V} \times [0, l]}: \bar{V} \times [0, l] \rightarrow M$ fulfils the conditions given in Definition 1.2.4.

I think this should work... I think it would be nicer to do this directly in Lean and then later write up the maths proof. $\square$

which will then allow us to derive

Theorem 1.2.15 (Collar neighbourhood theorem). *Let M be a C^k -manifold with boundary, then ∂M is collared.*

Bibliography

- [Con71] Robert Connelly. “A New Proof of Brown’s Collaring Theorem”. In: *Proceedings of the American Mathematical Society* 27.1 (1971), pp. 180–182. ISSN: 00029939, 10886826. URL: <http://www.jstor.org/stable/2037284> (visited on 04/29/2026).